Complete monotonicity of Tsallis entropy under heat flow: a sharp order classification

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Abstract

For a probability density evolving under the linear heat equation, consider the alternating signs of all derivatives of its Tsallis entropy. We prove that the quadratic order $\alpha = 2$ is the unique positive order for which these inequalities hold universally. For each fixed $\alpha > 0$ with $\alpha \notin \{1, 2\}$, we give an explicit family of smooth, strictly positive, strictly log-concave densities $f_{\alpha,m}$ with Gaussian tails, indexed by the derivative order m, for which the signed mth entropy derivative is negative for every sufficiently large m. The input density varies with m. Previously established Shannon counterexamples supply the case $\alpha = 1$. The proof extends the two-frequency spectral construction and Gaussian localisation of Zou, Fan, Gao and Wang: an order-dependent choice of the second Fourier amplitude makes the unique dominant spectral coefficient equal to $-\alpha|\alpha - 2|/8$. A separate example at $\alpha = 3/2$ specifies a single real-line density and certifies failure at derivative order 13 using an exact rational periodic calculation and a quantitative analytic transfer. The result classifies the order parameter pointwise; it does not claim the first refutation of the Shannon conjecture or a new localisation mechanism.

1 Question, conventions and scope

Adding Gaussian noise smooths a probability density. Entropy inequalities ask which features of that smoothing can be described by universal signs. An increasing entropy curve is one such property; alternating signs at every derivative order are a much stronger requirement. This paper determines precisely which positive Tsallis orders can satisfy that whole hierarchy for every input density.

Let

$$P_t = e^{t\Delta/2}, \qquad t \geq 0,$$

be the heat semigroup on $\mathbb{R}^d$. Thus $P_t f$ is the density obtained by adding centred Gaussian noise with covariance tI_d. For $\alpha > 0$, $\alpha \neq 1$, define

$$J_{\alpha,f}(t) = \int_{\mathbb{R}^d} (P_t f)^\alpha \, \mathrm{d}x, \qquad S_{\alpha,f}(t) = \frac{1 - J_{\alpha,f}(t)}{\alpha - 1}. \tag{1}$$

At $\alpha = 1$, use the continuous extension

$$S_{1,f}(t) = -\int_{\mathbb{R}^d} (P_t f) \log(P_t f) \, \mathrm{d}x.$$

The property at issue is

$$(-1)^{k-1} S^{(k)}_{\alpha,f}(t) \geq 0 \quad \text{for every } k \geq 1 \text{ and } t > 0. \tag{2}$$

The first inequality says that entropy increases under smoothing; the second says that its graph is concave. Higher inequalities impose additional restrictions that do not follow from these two. Failure of the full hierarchy therefore does not by itself refute an established low-order sign theorem. Equivalently, $S'_{\alpha,f}$ is completely monotone. Entropy itself need not be nonnegative, so we state the derivative inequalities directly. A derivative assertion is understood where the relevant entropy is finite and the derivative exists; every density constructed here has Gaussian tails and an entropy profile analytic in a neighbourhood of the time being used.

Wu, Yu and Guo [1] conjectured (2) for every $\alpha \in (1, 2)$, under this heat convention, and proved the quadratic case. Their conjecture implies the corresponding Shannon conjecture as $\alpha \to 1$. Gu and Sellke [2] subsequently disproved the Shannon conjecture by a certified finite atomic example and also proved the existence of log-concave counterexamples. Zou, Fan, Gao and Wang [3] then gave an explicit smooth strictly log-concave Shannon family, using two interacting Fourier modes localised by a Gaussian envelope.

Earlier results also delimit the scope of the question. Wu, Yu and Guo [1, Example 2] reported failures for the pairs $(k, \alpha) = (3, 33), (4, 8), (5, 5), (6, 4), (9, 3)$ using a symmetric two-point input. Wu and Yu [4, Section III-B and Appendix B] established related low-order signs and conditional implications, and gave the quadratic identity in arbitrary dimension. Sun [5] proved concavity of Tsallis entropy for $\alpha \in (0, 3]$ in dimension one and for $\alpha \in [1, 3]$ in higher dimensions. These lower-order conclusions are consistent with the higher-derivative failures proved here.

These results already preclude the entire interval conjecture from being true: strict failure for Shannon entropy and continuity in α produce failures at some orders close to 1. The question treated here is sharper: *which individual positive orders can satisfy (2) universally?* Our answer is that only $\alpha = 2$ survives, even if the universal quantifier is restricted to smooth strictly positive strictly log-concave densities with Gaussian tails.

The mechanism of the proof is adapted directly from [3], and we credit that source throughout. The additional steps are the explicit tuning of the second Fourier amplitude for an arbitrary fixed Tsallis order, and a transfer estimate for the power moment, whose Gaussian scaling factor is absent from the Shannon periodic term. The certified finite example is a separate concrete witness; it is not a claim about the least possible derivative order.

Theorem 1 (Sharp universal classification). Fix a dimension $d \geq 1$. The only $\alpha > 0$ for which (2) holds for every smooth strictly positive strictly log-concave probability density with Gaussian tails on $\mathbb{R}^d$ is $\alpha = 2$.

For $\alpha = 2$, the inequalities hold, more generally, for every probability density and every positive time. For each $\alpha \neq 2$, a density in the stated smaller class violates a strict inequality at a positive time.

The construction for $\alpha \notin \{1, 2\}$ is explicit. Fix such an α, and set

$$P_\alpha = (\alpha - 3)(\alpha - 4)(\alpha - 5), \qquad B_\alpha = \operatorname{sgn}(2 - \alpha) - \frac{P_\alpha}{288}. \tag{3}$$

For integers $m \geq 1$, put

$$a_m = 2^{-m/4}, \qquad b_m = B_\alpha 2^{-m} = B_\alpha a_m^4, \qquad \kappa_m = 8a_m, \tag{4}$$

and, whenever the periodic factor is positive, define

$$f_{\alpha,m}(x) = Z_{\alpha,m}^{-1} e^{-\kappa_m x^2/2}(1 + a_m \cos x + b_m \cos 2x), \qquad x \in \mathbb{R}. \tag{5}$$

The exact normalising constant is

$$Z_{\alpha,m} = \sqrt{\frac{2\pi}{\kappa_m}}\, \Xi_{\alpha,m}, \qquad \Xi_{\alpha,m} = 1 + a_m e^{-1/(2\kappa_m)} + b_m e^{-2/\kappa_m}. \tag{6}$$

The index m specifies both the density and the derivative order.

Theorem 2 (Explicit log-concave family). Fix $\alpha > 0$, $\alpha \notin \{1, 2\}$, and let (3)–(6) hold. For every sufficiently large m, $f_{\alpha,m}$ is a smooth strictly positive strictly log-concave probability density with Gaussian tails. Define

$$\beta = \frac{3}{2\sqrt{2}} > 1, \qquad C_{\alpha,m} = \alpha^{-1/2} \left(\frac{\kappa_m}{2\pi}\right)^{(\alpha-1)/2} \Xi_{\alpha,m}^{-\alpha} > 0. \tag{7}$$

Then

$$\frac{(-1)^{m-1} S^{(m)}_{\alpha, f_{\alpha,m}}(0)}{C_{\alpha,m}} = -\frac{\alpha|\alpha - 2|}{8}\, \beta^m + o_\alpha(\beta^m) \qquad (m \to \infty). \tag{8}$$

In particular, for each sufficiently large m, the signed derivative is negative at zero and at every sufficiently small positive time.

All asymptotic statements in this theorem hold for a *fixed* α. No uniformity as $\alpha \to 1$, $\alpha \to 2$, $\alpha \to 0$ or $\alpha \to \infty$ is asserted.

1.1 Witness map and proof roadmap

Construction	Entropy order and derivative	Shape and scope
Theorem 2	Each fixed $\alpha \notin \{1, 2\}$; all sufficiently large m	A different smooth strictly log-concave density for each m; threshold not quantified
Inherited Shannon theorem	$\alpha = 1$; all sufficiently large m	The cited log-concave family, with density depending on m
Theorem 7	$\alpha = 3/2$, derivative 13, time $1/50$	One specified smooth positive density; no log-concavity or least-order claim

The periodic calculation first arranges a negative coefficient at one spectral level. Repeated differentiation amplifies that level faster than all others. Gaussian localisation then turns the periodic perturbation into an integrable probability density, while controlling the errors even after the large number of derivatives. The quadratic case is different: its power integral is a nonnegative Fourier-weighted mixture of decaying exponentials. The finite witness later replaces an unspecified asymptotic threshold by exact rational bounds for one order and time.

2 The periodic calculation

For a periodic integrable function, write

$$\langle F \rangle_{\mathbb{T}} = \frac{1}{2\pi} \int_0^{2\pi} F(x)\, \mathrm{d}x.$$

Define

$$r_m(s, x) = 1 + a_m e^{-s/2} \cos x + b_m e^{-2s} \cos 2x, \qquad D_{\alpha,m}(s) = \frac{\langle r_m(s, \cdot)^\alpha \rangle_{\mathbb{T}} - 1}{\alpha - 1}. \tag{9}$$

This is the periodic part of the negative Tsallis entropy. The heat exponents and the choice of amplitudes in (4) follow the spectral construction in [3].

Lemma 3 (Periodic asymptotic). For every fixed $\alpha > 0$, $\alpha \notin \{1, 2\}$,

$$(-1)^m D^{(m)}_{\alpha,m}(0) = -\frac{\alpha|\alpha - 2|}{8}\, \beta^m + O_\alpha(1). \tag{10}$$

Moreover, the absolute sum of the differentiated monomial expansion is $O_\alpha(\beta^m)$.

Proof. For large m, $a_m + |b_m| < 1/2$. Expanding the analytic function $(1 + u)^\alpha$ around $u = 0$, using that the linear term has zero circle mean, gives

$$D_{\alpha,m}(s) = \sum_{\substack{p,q \geq 0 \\ p+q \geq 2}} \Gamma_{p,q}(\alpha)\, a_m^p b_m^q e^{-(p+4q)s/2}, \tag{11}$$

where

$$\Gamma_{p,q}(\alpha) = \frac{\binom{\alpha}{p+q}}{\alpha - 1} \binom{p+q}{p} \langle \cos^p x \cos^q 2x \rangle_{\mathbb{T}}\,. \tag{12}$$

For each fixed m, this series can be differentiated m times near zero: at total degree $\ell = p + q$, the time factor contributes at most $(2\ell)^m$, whereas the remaining absolute sum is bounded by a fixed geometric factor times a binomial-series coefficient. Taking m large enough that $a_m + |b_m|$ is smaller than a fixed analytic radius yields an absolutely convergent differentiated series.

The substitution $x \mapsto \pi - x$ shows that the trigonometric average in (12) vanishes for odd p. Write $p = 2h$ and set $j = h + 2q$. Then

$$\frac{p + 4q}{2} = j, \qquad a_m^{2h} b_m^q = B_\alpha^q 2^{-mj/2}.$$

Consequently,

$$(-1)^m D_{\alpha,m}^{(m)}(0) = \sum_{j \geq 1} A_j(\alpha) \beta_j^m, \qquad \beta_j = j 2^{-j/2}, \tag{13}$$

where

$$A_j(\alpha) = \sum_{\substack{h,q \geq 0 \\ h+2q=j \\ 2h+q \geq 2}} \Gamma_{2h,q}(\alpha) B_\alpha^q.$$

The ratio $\beta_{j+1}/\beta_j = (j+1)/(j\sqrt{2})$ shows that $j = 3$ is the unique maximum. Specifically, $\beta_2 = \beta_4 = 1$, $\beta_3 = \beta$, and $\beta_j < 1$ for every $j \notin \{2, 3, 4\}$.

There are exactly two pairs at level $j = 3$: $(p, q) = (2, 1)$ and $(6, 0)$. Since

$$\left\langle \cos^2 x \cos 2x \right\rangle_{\mathbb{T}} = \frac{1}{4}, \qquad \left\langle \cos^6 x \right\rangle_{\mathbb{T}} = \frac{5}{16},$$

their sum is

$$\begin{aligned} A_3(\alpha) &= \frac{3}{4} \frac{\binom{\alpha}{3}}{\alpha - 1} B_\alpha + \frac{5}{16} \frac{\binom{\alpha}{6}}{\alpha - 1} \\ &= \frac{\alpha(\alpha - 2)}{8} B_\alpha + \frac{\alpha(\alpha - 2)(\alpha - 3)(\alpha - 4)(\alpha - 5)}{2304} = -\frac{\alpha |\alpha - 2|}{8}. \end{aligned} \tag{14}$$

This is the only order-dependent tuning needed to obtain a negative leading coefficient.

For completeness, we control the infinite sum. Analyticity of

$$\frac{(1+u)^\alpha - 1 - \alpha u}{\alpha - 1}$$

on $|u|<1$ implies, by a Cauchy estimate on $|u|=1/2$, that

$$\left|\frac{\binom{\alpha}{\ell}}{\alpha-1}\right| \le K_\alpha 2^\ell, \qquad \ell \ge 2.$$

At level j there are at most $j+1$ pairs and $\ell = 2j-3q \le 2j$. Therefore the sum $\mathcal{A}_j$ of the absolute monomial coefficients at that level satisfies

$$|A_j| \le \mathcal{A}_j \le K_\alpha (j+1) R_\alpha^j, \qquad R_\alpha = 16\max\{1,|B_\alpha|\}^2. \tag{15}$$

For $j \ge 16$, one has $j \le 2^{j/4}$, hence $\beta_j \le 2^{-j/4}$. Choose an integer m_0 with $R_\alpha 2^{-m_0/4} < 1$. Then

$$\sum_{j\ge 16} \mathcal{A}_j \beta_j^m \le K_\alpha \sum_{j\ge 16} (j+1)(R_\alpha 2^{-m_0/4})^j < \infty, \qquad m \ge m_0,$$

uniformly in m. The finitely many levels $j \le 15$, $j \ne 3$, contribute $O_\alpha(1)$. Equation (14) proves (10); retaining absolute values proves the last assertion. □

3 Gaussian localisation for power moments

We now adapt the exact localisation method in [3] to $J_{\alpha,f}$. Throughout this section, $\alpha > 0$, $\alpha \notin \{1,2\}$, is fixed, and the subscript m is sometimes suppressed.

3.1 Positivity and strict log-concavity

For sufficiently large m, $a_m + |b_m| \le 1/2$ and $|b_m| \le a_m/8$. A simple sufficient condition is $a_m \le 1/4$ together with $|B_\alpha| a_m^3 \le 1/8$. If

$$\rho_m(x) = 1 + a_m \cos x + b_m \cos 2x,$$

then $\rho_m \ge 1/2$ and $|\rho_m''| \le a_m + 4|b_m| \le 3a_m/2$. Therefore

$$(\log \rho_m)'' = \frac{\rho_m''}{\rho_m} - \frac{(\rho_m')^2}{\rho_m^2} \le 3a_m,$$

and

$$(\log f_{\alpha,m})'' = -\kappa_m + (\log \rho_m)'' \le -5a_m < 0. \tag{16}$$

The remaining regularity and tail properties follow immediately from (5) and the bounds on ρ_m.

3.2 Exact heat evolution

For $\kappa > 0$ and $n \in \mathbb{Z}$, completing the square in the Gaussian convolution gives the identity from [3, Lemma 3]:

$$P_t\big(e^{-\kappa x^2/2}e^{inx}\big) = (1+\kappa t)^{-1/2}e^{-\kappa x^2/(2(1+\kappa t))}e^{-n^2 t/(2(1+\kappa t))}e^{inx/(1+\kappa t)}. \tag{17}$$

Indeed, the exponent in the convolution integrand is

$$-\frac{1+\kappa t}{2t}\left(y - \frac{x+int}{1+\kappa t}\right)^2 - \frac{\kappa x^2}{2(1+\kappa t)} - \frac{n^2 t}{2(1+\kappa t)} + \frac{inx}{1+\kappa t},$$

and Gaussian integration proves (17) for $t > 0$, with continuity at zero.

Set

$$d_m(t) = 1+\kappa_m t, \qquad \tau_m(t) = \frac{t}{d_m(t)}, \qquad \lambda_m(t) = \alpha\kappa_m d_m(t), \qquad \nu = \frac{1-\alpha}{2}. \tag{18}$$

Applying (17) to the three modes in (5) yields

$$P_t f_{\alpha,m}(x) = Z_{\alpha,m}^{-1} d_m(t)^{-1/2} e^{-\kappa_m x^2/(2d_m(t))} r_m\left(\tau_m(t), \frac{x}{d_m(t)}\right). \tag{19}$$

For real $\lambda > 0$, let $\langle\cdot\rangle_\lambda$ denote expectation under the centred Gaussian law of precision λ. Changing variables $x = d_m(t)y$ in the power integral gives the scalar identity

$$Z_{\alpha,m}^{-\alpha} d_m(t)^{1-\alpha/2}\sqrt{\frac{2\pi}{\alpha\kappa_m d_m(t)}} = C_{\alpha,m} d_m(t)^{(1-\alpha)/2}.$$

Consequently,

$$J_{\alpha,f_{\alpha,m}}(t) = C_{\alpha,m} d_m(t)^{\nu} \left\langle r_m(\tau_m(t),\cdot)^{\alpha}\right\rangle_{\lambda_m(t)}. \tag{20}$$

The constant $C_{\alpha,m}$ is exactly (7); no asymptotic normalisation is being substituted here.

Lemma 4 (Power-moment transfer). As $m \to \infty$ with α fixed,

$$\frac{(-1)^m J^{(m)}_{\alpha,f_{\alpha,m}}(0)}{C_{\alpha,m}(\alpha-1)} = (-1)^m D^{(m)}_{\alpha,m}(0) + o_\alpha(\beta^m). \tag{21}$$

The profile in (20) has a holomorphic extension to a neighbourhood of zero for all sufficiently large m.

Proof. We separate Fourier filtering, the scalar Gaussian factor, and the time change. A broad Gaussian average suppresses nonconstant Fourier modes. The power moment also introduces a scalar Gaussian factor, absent from the Shannon periodic term, and a small reparametrisation of time. Each contribution must remain smaller than the negative β^m scale after taking m derivatives.

Fourier filtering. Use the complex time disk $|t| \le 1/2$ and the spatial strip $|\operatorname{Im} y| \le 1$. For large m, $|\kappa_m t| \le 1/4$, $|\tau_m(t)| \le 1$, and

$$\left| a_m e^{-\tau_m(t)/2} \cos y + b_m e^{-2\tau_m(t)} \cos 2y \right| < \frac{1}{2}$$

on this product domain. Consequently

$$G_m(t, y) = r_m(\tau_m(t), y)^\alpha$$

is defined there by the principal power and is holomorphic and uniformly bounded, with a bound depending only on α. Its Fourier coefficients satisfy

$$|\widehat{G}_m(n, t)| \le K_\alpha e^{-|n|}. \tag{22}$$

For complex λ with positive real part, the Gaussian bracket has a holomorphic extension, using the principal square root, and

$$\left\langle e^{iny} \right\rangle_\lambda = e^{-n^2/(2\lambda)}.$$

Since $|d_m(t) - 1| \le 1/4$,

$$\operatorname{Re} \frac{1}{\lambda_m(t)} = \frac{1}{\alpha \kappa_m} \operatorname{Re} \frac{1}{d_m(t)} \ge \frac{1}{3\alpha\kappa_m}.$$

Normal convergence of the Fourier series permits termwise integration. Equations (22) and the preceding estimate imply

$$\left| \langle G_m(t, \cdot) \rangle_{\lambda_m(t)} - \langle G_m(t, \cdot) \rangle_{\mathbb{T}} \right| \le K_\alpha e^{-c_\alpha/\kappa_m} \qquad (|t| \le 1/2), \tag{23}$$

where one may take $c_\alpha = 1/(6\alpha)$. The same estimate, with another constant, holds after multiplying by $d_m(t)^\nu/(\alpha - 1)$. Cauchy's estimate bounds the resulting mth derivative at zero by

$$K_\alpha 2^m m! e^{-c_\alpha/\kappa_m} = o_\alpha(\beta^m),$$

because $1/\kappa_m = 2^{m/4}/8$. This construction also supplies the stated holomorphic extension, agreeing with (20) on real nonnegative times.

The constant periodic mode. The zero Fourier coefficient is

$$\langle G_m(t, \cdot) \rangle_{\mathbb{T}} = 1 + (\alpha - 1) D_{\alpha,m}(\tau_m(t)).$$

Thus, after division by $\alpha - 1$, the constant term contributes $d_m(t)^\nu/(\alpha - 1)$. Its mth derivative has absolute value at most

$$\frac{(|\nu| + m)^m \kappa_m^m}{|\alpha - 1|} = \exp\{-\tfrac{1}{4}(\log 2) m^2 + O_\alpha(m \log m)\} = o_\alpha(\beta^m).$$

Time change and the remaining scalar factor. We use a uniform estimate for each spectral mode. For $L \geq 1$ and integer $n \geq 1$, define

$$T_n(L,\kappa) = (-1)^n \left.\frac{\mathrm{d}^n}{\mathrm{d}t^n} e^{-Lt/(1+\kappa t)}\right|_{t=0}.$$

Expanding the exponential and then $(1+\kappa t)^{-\ell}$ yields

$$T_n(L,\kappa) = \sum_{r=0}^{n-1} \frac{n!}{(n-r)!}\binom{n-1}{r}\kappa^r L^{n-r}, \qquad T_0(L,\kappa) = 1. \tag{24}$$

This identity and the associated estimate are the time-change calculation of [3]. Every term is nonnegative, and

$$1 \leq T_n(L,\kappa)/L^n \leq e^{n^2\kappa/L} \quad (n \geq 0), \tag{25}$$

where the lower bound for $n = 0$ is an equality.

For the extra factor required by the power moment, put

$$U_m(L,\kappa) = (-1)^m \left.\frac{\mathrm{d}^m}{\mathrm{d}t^m}\left[(1+\kappa t)^\nu e^{-Lt/(1+\kappa t)}\right]\right|_{t=0}.$$

Leibniz' rule gives

$$U_m = \sum_{k=0}^{m}\binom{m}{k}(-1)^k(\nu)_{\underline{k}}\kappa^k T_{m-k}(L,\kappa).$$

Using $|(\nu)_{\underline{k}}| \leq (|\nu|+m)^k$, $\binom{m}{k} \leq m^k/k!$, and (25), we obtain

$$\left|\frac{U_m(L,\kappa)}{L^m} - 1\right| \leq \exp\left(\frac{(2m^2+|\nu|m)\kappa}{L}\right) - 1. \tag{26}$$

For clarity, the $k = 0$ term contributes at most $e^{m^2\kappa/L} - 1$ to this difference; the sum over $k \geq 1$ is at most

$$e^{m^2\kappa/L}\left(e^{m(|\nu|+m)\kappa/L} - 1\right),$$

whose sum is (26).

For $\kappa = \kappa_m$, the right side of (26) is $o(1)$ uniformly in $L \geq 1$. The periodic expansion can be written

$$D_{\alpha,m}(s) = \sum_{j\geq 1} A_j(\alpha) 2^{-mj/2} e^{-js}.$$

Apply (26) with $L = j$ and use the absolute-sum estimate in Lemma 3. It follows that

$$(-1)^m \left.\frac{\mathrm{d}^m}{\mathrm{d}t^m}[d_m(t)^\nu D_{\alpha,m}(\tau_m(t))]\right|_{t=0} = (-1)^m D^{(m)}_{\alpha,m}(0) + o_\alpha(\beta^m).$$

Termwise differentiation is justified by the same absolutely convergent monomial series as in Lemma 3, or by (15) and (26). Combining the three estimates proves (21). □

Proof of Theorem 2. Positivity, Gaussian tails, smoothness and strict log-concavity were proved in (16). From (1),

$$(-1)^{m-1} S^{(m)}_{\alpha,f}(t) = \frac{(-1)^m J^{(m)}_{\alpha,f}(t)}{\alpha - 1}.$$

Combine Lemmas 3 and 4 to obtain (8). Its leading coefficient is strictly negative. The holomorphic extension in Lemma 4 gives continuity of the mth derivative at zero, so the strict sign persists for a sufficiently small interval of positive times. □

4 The quadratic case, Shannon order and higher dimensions

Proposition 5 (The quadratic case). For any probability density f on $\mathbb{R}^d$, $t > 0$ and integer $k \geq 1$,

$$(-1)^{k-1} S^{(k)}_{2,f}(t) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} |\xi|^{2k} |\widehat{f}(\xi)|^2 e^{-t|\xi|^2} \,\mathrm{d}\xi \geq 0. \tag{27}$$

In dimension one this equals $\int_{\mathbb{R}} |\partial_x^k P_t f|^2 \,\mathrm{d}x$.

Proof. Use the Fourier convention $\widehat{f}(\xi) = \int e^{-ix\cdot\xi} f(x) \,\mathrm{d}x$. Since $|\widehat{f}| \leq 1$ and $\widehat{P_t f} = e^{-t|\xi|^2/2} \widehat{f}$, Plancherel's identity gives

$$J_{2,f}(t) = \frac{1}{(2\pi)^d} \int |\widehat{f}(\xi)|^2 e^{-t|\xi|^2} \,\mathrm{d}\xi.$$

Gaussian domination permits differentiation of every order at positive times. Since $S_{2,f} = 1 - J_{2,f}$, (27) follows. This reproves the quadratic fact in [1] and its arbitrary-dimensional version in [4, Appendix B]. □

At the Shannon order, we use the already established theorem of Zou et al. [3, Theorem 1]. With $a_m = 2^{-m/4}$, $b_m = 2^{-m}$ and $\kappa_m = 8a_m$, their density (5) is smooth, strictly positive and strictly log-concave for $m \geq 5$, and

$$(-1)^{m-1} S^{(m)}_{1,f_m}(0) = -\frac{11}{96}\beta^m + o(\beta^m). \tag{28}$$

Their analytic construction also gives failure at small positive times. Equation (28) is quoted from prior work, not asserted as a new case of our result. Together with Theorem 2 and Proposition 5, it proves Theorem 1 in dimension one.

Corollary 6 (Every dimension). For every $d \geq 2$ and $\alpha > 0$, $\alpha \neq 2$, there is a smooth strictly positive strictly log-concave probability density with Gaussian tails on $\mathbb{R}^d$ for which (2) fails.

Proof. First take a one-dimensional density f, a derivative order k, and a positive time t_0 with a strict negative signed derivative, as obtained above. Let g_R be the centred Gaussian density of covariance RI_{d-1} and use $f \otimes g_R$. Strict log-concavity and the other stated regularity properties are preserved.

For $\alpha \neq 1$, the power moments factor, and

$$J_{\alpha,g_R}(t) = K_{\alpha,R,d}(1 + t/R)^{-(d-1)(\alpha-1)/2},$$

where $K_{\alpha,R,d} > 0$ is independent of time. The factor following $K_{\alpha,R,d}$ tends to 1 in C^k on every compact time interval as $R \to \infty$. Consequently the signed kth derivative of $S_{\alpha,f\otimes g_R}$, after division by $K_{\alpha,R,d}$, converges at t_0 to the strictly negative signed derivative for f. It is therefore negative for sufficiently large R.

For $\alpha = 1$, Shannon entropy is additive, and the Gaussian summand contributes

$$(-1)^{k-1} S^{(k)}_{1,g_R}(t_0) = \frac{d-1}{2} \frac{(k-1)!}{(R+t_0)^k} \longrightarrow 0.$$

Thus the strict negative sign is again preserved for large R. Combining this corollary with Proposition 5 completes Theorem 1 in every dimension. □

5 A completely specified witness at order $\alpha = 3/2$

The asymptotic family proves the classification and the log-concave strengthening, but it does not give a numerical threshold for m at each α. The following separate construction gives a finite certificate at derivative order 13 for the open-interval order $\alpha = 3/2$.

Theorem 7 (Finite real-line witness). Let $R = 2^{40}$ and let γ_R denote the centred Gaussian density of variance R. Set

$$\begin{aligned} f_R(x) &= \frac{\gamma_R(x)}{Z_R}\left(1 + \frac{1}{10}\cos x + \frac{1}{32000}\cos 2x\right), \\ Z_R &= 1 + \frac{1}{10}e^{-R/2} + \frac{1}{32000}e^{-2R}. \end{aligned} \tag{29}$$

This is a smooth strictly positive probability density with Gaussian tails. Under the convention $P_t = e^{t\Delta/2}$, put

$$C_R = (3/2)^{-1/2}(2\pi R)^{-1/4}.$$

Then

$$S^{(13)}_{3/2,f_R}(1/50) < -\frac{C_R}{1024} < 0. \tag{30}$$

Since $(-1)^{12} = 1$, this violates (2). No log-concavity assertion is made for this particular finite witness.

Proof. For the finite calculation only, write $Q_t = e^{t\Delta} = P_{2t}$. On the circle define

$$v(t,x) = 1 + ae^{-t}\cos x + be^{-4t}\cos 2x, \qquad a = \frac{1}{10}, \quad b = \frac{1}{32000}, \quad F(t) = \left\langle v(t,\cdot)^{3/2} \right\rangle_{\mathbb{T}}.$$

The exact rational certificate in Appendix A proves

$$(-1)^{13} F^{(13)}(1/100) < -6. \tag{31}$$

We give all analytic estimates needed to transfer this finite sign to (29).

Put $M_R(t) = \int (Q_t f_R)^{3/2}\,\mathrm{d}x$, $d = 1 + 2t/R$ and $\tau = t/d$. The exact window identity gives

$$Q_t f_R(x) = Z_R^{-1}\gamma_{Rd}(x)[1 + ae^{-\tau}\cos(x/d) + be^{-4\tau}\cos(2x/d)].$$

Let $\widehat{V}_\ell(s)$ be the Fourier coefficients in x of $v(s,x)^{3/2}$. Gaussian integration gives

$$G_R(t) := \frac{M_R(t)}{C_R} = Z_R^{-3/2} d^{-1/4} \sum_{\ell\in\mathbb{Z}} \widehat{V}_\ell(\tau) e^{-\ell^2 R/(3d)}. \tag{32}$$

In particular, $\widehat{V}_0 = F$. The normally convergent Fourier expression on the right, rather than a complex power of the real-line density, supplies the holomorphic extension used below.

Let $T = 1/100$ and use the disk $|z - T| \le 1$. The following bounds hold for every $R \ge 1024$, hence for the specified R. We have $|z| < 11/10$,

$$|d - 1| \le \frac{11}{5R}, \qquad \mathrm{Re}(1/d) > \frac{1}{2}, \qquad |\tau - z| \le \frac{121}{50R(1 - 11/(5R))} < \frac{5}{2R}.$$

Every point on the segment from z to τ has real part greater than $-11/10$. For $\mathrm{Re}\, s \ge -11/10$ and $|\,\mathrm{Im}\, x| \le 1$, use

$$e^{11/10} < 10/3, \qquad e^{22/5} < 135, \qquad \cosh 1 < 2, \qquad \cosh 2 < 4.$$

For example, the exponential bounds follow from $e < 3$, $e^{1/10} < 10/9$ and $e^{2/5} < 5/3$, the last two being immediate geometric majorants of the exponential series. Hence

$$|ae^{-s}\cos x + be^{-4s}\cos 2x| < \frac{2}{3} + \frac{27}{1600} < \frac{3}{4}.$$

The principal $3/2$ power is therefore holomorphic on this domain, has modulus less than 3, and obeys

$$|\widehat{V}_\ell(s)| \le 3e^{-|\ell|}.$$

Also $|F| < 3$, and differentiation under the circle average gives

$$|F'(s)| \le \frac{3}{2}\sqrt{\frac{7}{4}}\left(\frac{2}{3} + \frac{27}{400}\right) < 2.$$

Consequently $|F(\tau) - F(z)| < 5/R$. The nonzero Fourier modes in (32) satisfy

$$\left|\sum_{\ell \neq 0} \widehat{V}_\ell(\tau) e^{-\ell^2 R/(3d)}\right| < 6e^{-R/6}. \tag{33}$$

Integration of the derivative of $w^{-1/4}$ along the segment from 1 to d gives

$$|d^{-1/4} - 1| < \frac{11}{10R}, \qquad |d^{-1/4}| < 2.$$

Furthermore, $Z_R \ge 1$ and

$$Z_R - 1 < \frac{11}{100} e^{-R/2} \le \frac{11}{100R}.$$

Since $|Z_R^{-3/2} - 1| \le (3/2)(Z_R - 1)$, the factor $A = Z_R^{-3/2} d^{-1/4}$ obeys

$$|A - 1| < \frac{2}{R}, \qquad |A| < 2.$$

For $R \ge 1024$, $12e^{-R/6} < 1/R$: the positive exponential series gives $e^{R/6} > R^3/1296$, and $R^2 > 15552$. Writing

$$AF(\tau) - F(z) = [F(\tau) - F(z)] + (A - 1)F(\tau)$$

and using (33), the time-change, scalar-factor and nonzero-mode errors are respectively smaller than $5/R$, $6/R$ and $1/R$. We may therefore use the generous bound

$$\sup_{|z-T| \le 1} |G_R(z) - F(z)| < 100/R. \tag{34}$$

The Fourier series in (32) converges normally on this disk, so it defines a holomorphic function agreeing with M_R/C_R at real nonnegative times. Cauchy's estimate gives

$$|G_R^{(13)}(T) - F^{(13)}(T)| < \frac{100\,13!}{2^{40}} = \frac{152026875}{268435456} < 1. \tag{35}$$

The final comparison is also checked in exact integer arithmetic by the accompanying certificate.

Equation (31) says $F^{(13)}(T) > 6$, so (35) implies $M_R^{(13)}(T)/C_R > 5$. Finally, $P_t = Q_{t/2}$, $t = 2T = 1/50$, and $S_{3/2,f} = 2(1 - J_{3/2,f})$. Thus

$$J_{3/2,f_R}^{(13)}(1/50) > \frac{5C_R}{8192}, \qquad S_{3/2,f_R}^{(13)}(1/50) < -\frac{5C_R}{4096} < -\frac{C_R}{1024}.$$

This proves the claim. □

6 Dilation and the scale of a counterexample

Corollary 8 (Elementary dilation consequences). Let f be a one-dimensional density with finite positive variance, and suppose its signed entropy derivative of order $k \geq 1$ is strictly negative at a positive time t_0. For $a > 0$ define $f_a(x) = a^{-1} f(x/a)$. Then the same signed derivative for f_a is strictly negative at $a^2 t_0$. Consequently a witness can be rescaled to variance one, or it can be rescaled to have failure at any prescribed positive time. These are separate choices of the single scale parameter.

Proof. Gaussian convolution and the substitution $x = ay$ give

$$P_t f_a(x) = a^{-1}(P_{t/a^2} f)(x/a).$$

For $\alpha \neq 1$ this implies

$$J_{\alpha,f_a}(t) = a^{1-\alpha} J_{\alpha,f}(t/a^2),$$
$$(-1)^{k-1} S^{(k)}_{\alpha,f_a}(t) = a^{1-\alpha-2k}(-1)^{k-1} S^{(k)}_{\alpha,f}(t/a^2).$$

The multiplicative factor is positive. For $\alpha = 1$, $S_{1,f_a}(t) = S_{1,f}(t/a^2) + \log a$, so the derivative factor is a^{-2k}, again positive. Variance transforms as $\mathrm{Var}(f_a) = a^2 \mathrm{Var}(f)$. Thus $a = \mathrm{Var}(f)^{-1/2}$ gives variance one, whereas $a = (T/t_0)^{1/2}$ places failure at a prescribed $T > 0$.

Smoothness, strict positivity and Gaussian tails are preserved by dilation. If f is strictly log-concave, then $\log f_a(x) = \log f(x/a) - \log a$ is strictly concave by affine composition. For the constructed family its negative curvature transforms by $(\log f_a)''(x) = a^{-2}(\log f)''(x/a)$. The constructed densities have finite positive variance. This proves the conclusions. □

This corollary is a standard heat-scaling consequence, not a new localisation mechanism. The ratio of failure time to variance is unchanged: the two quantities cannot be prescribed independently. Dilation does not supply a log-concavity claim for the finite witness and does not establish a uniform curvature bound across the asymptotic family.

7 What the classification establishes

Theorem 1 settles the universal order question, including each individual order in $(1, 2)$. The exception at $\alpha = 2$ reflects the quadratic Fourier representation (27). For $\alpha \notin \{1, 2\}$, the power-entropy expansion has a nonzero cubic contribution; the second harmonic can be chosen to make the combined dominant spectral coefficient negative. The Shannon case is supplied separately by its cited entropy expansion. The sixth-order contribution at the same spectral level is retained exactly in (14), rather than discarded as a lower-order perturbation.

The bounds are pointwise in α. In the displayed family, the sufficient curvature lower bound $5a_m$ tends to zero; no uniform curvature or variance bound across that family is claimed. Dilation gives a different normalization of each witness, not a freely chosen physical noise scale.

The result does not determine the first derivative order at which a particular α admits failure. The asymptotic family establishes existence with strong regularity and shape restrictions, while the order-13 witness gives one reproducible finite example. Neither construction estimates a physically typical noise level. The method and its limitations are inherited in substantial part from the Shannon construction of [3]; the contribution here is the complete positive-order classification and the accompanying power-moment and finite-certificate details.

Appendix A. Exact rational circle certificate

This appendix explains the finite certificate used in (31). It involves no floating-point sign decision. Let

$$\mathcal{F}(z) = \left\langle (1 + az\cos x + bz^4 \cos 2x)^{3/2} \right\rangle_{\mathbb{T}} = \sum_{k\geq 0} c_k z^k, \qquad a = 1/10, \quad b = 1/32000.$$

Then $F(t) = \mathcal{F}(e^{-t})$. The coefficients are rational and can be computed as

$$c_k = \sum_{i+4j=k} \binom{3/2}{i+j} \binom{i+j}{i} a^i b^j \left\langle \cos^i x \cos^j 2x \right\rangle_{\mathbb{T}}. \tag{36}$$

The moments in this formula are exactly

$$\left\langle \cos^i x \cos^j 2x \right\rangle_{\mathbb{T}} = 2^{-(i+j)} \sum_{\substack{0\leq r\leq i,\, 0\leq s\leq j \\ i-2r+2(j-2s)=0}} \binom{i}{r}\binom{j}{s}.$$

Odd coefficients vanish by the substitution $x \mapsto \pi - x$. The script computes every coefficient through degree 23; the first four nonzero coefficients after $c_0 = 1$ are

$$c_2 = \frac{3}{1600}, \quad c_4 = \frac{9}{10240000}, \quad c_6 = -\frac{41}{3276800000}, \quad c_8 = \frac{12213}{83886080000000}.$$

At derivative order 13 and time $T = 1/100$,

$$(-1)^{13} F^{(13)}(T) = \sum_{k\geq 1} c_k k^{13} e^{-kT}. \tag{37}$$

For the infinite tail, use the analytic radius $\rho = 8$. Since

$$a\rho + b\rho^4 = \frac{116}{125} < 1, \qquad (1 + 116/125)^3 < 3^2,$$

the principal power is analytic for $|z| \leq 8$ and $|\mathcal{F}(z)| < 3$ there. Thus $|c_k| \leq 3\,8^{-k}$. The ratio of successive even Cauchy majorants for $k \geq 24$ is at most

$$q = \frac{(13/12)^{13}}{64} < 1.$$

Since $e^{-kT} \leq 1$, the absolute tail in (37) is bounded by

$$H = \frac{3\,24^{13}}{8^{24}(1-q)} < \frac{1}{1000}. \tag{38}$$

To bound the finite sum with rational arithmetic, bracket $E = e^{1/100}$ by

$$E_- = \sum_{j=0}^{12} \frac{(1/100)^j}{j!}, \qquad E_+ = E_- + \frac{(1/100)^{13}/13!}{1 - 1/1400}.$$

The omitted positive Taylor terms have successive ratio at most $1/1400$. Therefore the finite sum in (37) is at most

$$B = \sum_{\substack{2 \le k \le 22 \\ k \text{ even} \\ c_k \ge 0}} \frac{c_k k^{13}}{E_-^k} + \sum_{\substack{2 \le k \le 22 \\ k \text{ even} \\ c_k < 0}} \frac{c_k k^{13}}{E_+^k}.$$

The script checks the exact rational inequalities

$$B < -\frac{27}{4}, \qquad H < \frac{1}{1000}.$$

Consequently

$$(-1)^{13} F^{(13)}(1/100) \le B + H < -\frac{6749}{1000} < -6,$$

which proves (31). The readable diagnostic for $B + H$ is approximately -6.753909793468304; it is not used to decide the sign.

The file `checks/tsallis_order13_certificate.py` contains the exact certificate. It computes the coefficients, verifies the analytic-bound hypotheses and finite rational inequalities, and checks the integer comparison in (35) and the final entropy margin. The companion `checks/test_guards.py` checks intended failures. Each accepts `-output NEW_PATH.json` and refuses to overwrite an existing receipt. The mathematical certificate above is complete without the source listing. Executable acceptance uses explicit named guards rather than removable assertions; see `VERIFICATION.md` for the exact invocation, positive and intended-failure checks, interpreter modes and artifact identities. Arithmetic success verifies only the finite comparisons, not the analytic convolution, Fourier or Cauchy arguments.

Declarations

Data and code availability. This theoretical study uses no empirical dataset. The complete mathematical certificate is included above; the accompanying public package contains its executable checker and verification documentation.

AI assistance and research status. AI systems assisted the analysis, manuscript revision and internal checks. This is an unrefereed candidate, not external human peer review, unaffiliated reproduction or proof-assistant verification.

Contributions. Anonymous authorship is retained. The work comprises mathematical analysis, software, validation and writing; no individual human CRediT allocation is asserted from the available record.

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